Mean vs Median vs Mode

three averages, one winner each!

Quantitative Aptitude: Statistics

In simple words Mean shares equally, median is the middle one, mode is the most popular one.

1 Arithmetic Mean

Sum of all observations divided by the number of observations. It uses every value in the data.

Pulled by extreme values

2 Median and Mode

Median is the middle value of data arranged in order. Mode is the value that occurs most often.

Arrange data first for median

1. Mean uses every value, so one very large or small value drags it.
2. Median depends only on position, so extreme values do not disturb it.
3. Mode is the most frequent value; data can have no mode or more than one.
4. In a symmetrical distribution, mean = median = mode.



Swipe for the formula →

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The 3-Average Table

Mean vs Median vs Mode

pick the average like a pro

Empirical relation

$$\text{Mode} = 3 \times \text{Median} - 2 \times \text{Mean}$$

Holds for a moderately skewed distribution only. Use it to find the third average from two.

Average	Extreme values	Best used when
Mean	Affects it strongly	Data is even, all values matter
Median	Not affected	Skewed data, open-ended classes
Mode	Not affected	Most common item or size is needed

Same data, three stories! With an extreme value, the mean moves away while median and mode stay put.

Swipe for a memory trick ⟶

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Study Trick

Mean vs Median vs Mode

Remember M-M-M

Mean Meets all, Median is Middle, Mode is Most

Mean meets every value in the data. Median stands in the middle of the ordered list. Mode is the most popular value.

How to attempt it

1. Read which average is asked →
2. Arrange data in order (for median) →
3. Find the middle position or highest frequency →
4. Check for skewness or extreme values

Exam Tip

If the question says outliers or open-ended classes, choose median. If it says most common or most frequent, choose mode.

Doubt Buster

Can mean, median and mode all be different? Yes. In a skewed distribution they differ. They are equal only when the distribution is symmetrical.

Swipe for common mistakes →

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COMMON MISTAKES

Mean vs Median vs Mode

Don't lose easy marks!

✗ Taking the middle item without arranging the data in order

✓ Arrange data in ascending or descending order first, then locate the middle item.

✗ Saying median is badly affected by extreme values

✓ Median depends on position, so extreme values do not affect it. The mean is the one affected.

✗ Saying every data set has exactly one mode

✓ A data set can have no mode, one mode, or several modes.

✗ Using mean as the best average for open-ended classes

✓ Mean needs class mid-points, which open-ended classes lack. Median is preferred.

Save this for revision!

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